t3-w05 — condition C1

Three dimensional shapes II — Three dimensional figures: properties of cylinders and spheres, and volume

> The objective plus the lesson plan's design: position, method labels, materials, entry behaviour and assessment.

INPUT LAYERS

Input layer	What it carries	Channel	In this condition
Scope	what is to be taught, and to whom	prompt	**present**
Prior knowledge	what the learner already knew coming in	prompt	**present**
Materials	what the lesson was represented with	prompt	**present**
Position	where this objective sat in the sequence	prompt	**present**
Method	the shape of the teaching, as event labels	prompt	**present**
Enacted content	what was actually explained and worked through	prompt	absent
Set work	what the learners were asked to do in class	prompt	**present**
Assessment	the questions the lesson was checked with	prompt	absent
Figures	the drawings the lesson used	tool	absent

WHAT THE MODEL IS SENT

Contract `ComposeWriteObjectiveSection`, declaring: `artefact_spec, lesson_context, unit_context`.

Lesson context — shared by every section

`subject`

```
Mathematics
```

`grade_level`

```
JSS1
```

`topic`

```
Three dimensional shapes II
```

`subtopic`

```
Three dimensional figures: properties of cylinders and spheres, and volume
```

`term`

```
Third Term
```

`week`

```
5
```

`lesson_objectives`

• identify properties of cylinders and spheres
• find the volume of a cube and a cuboid

`entry_behaviour`

• properties of cubes and cuboid
• properties of pyramids and cones

`instructional_materials`

• shapes of cylinders and spheres.

Objective 1 — identify properties of cylinders and spheres

`objective_text`

identify properties of cylinders and spheres

`current_step`

1

`instructional_events`

• present_content
• elicit_performance
• present_content

`student_tasks`

• 1. How many circular faces does a solid cylinder have? 2. A cylinder has how many faces.

Withheld from this payload: `step_title, step_content, instructional_type, estimated_minutes, activities, assessment_points, source_evidence, source_figures, objective_learner_state` — carried by the plan or the unit, not sent under this condition.

Objective 2 — find the volume of a cube and a cuboid

`objective_text`

find the volume of a cube and a cuboid

`current_step`

2

`instructional_events`

- present_content
- provide_learning_guidance

`assessment_points`

- Calculate the volume of a cuboid 8cm long, 4cm wide and 3cm high.
- Calculate the volume of cuboid with dimensions 5m by 2m by 3m.
- What is the volume of a cube of 4cm $\times$ 4cm $\times$ 4cm.
- A rectangular tank hold 6m long by 2m wide holds 36m^{3} of water. How deep is the water in the tank.

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `student_tasks`, `activities`, `source_evidence`, `source_figures`, `objective_learner_state` — carried by the plan or the unit, not sent under this condition.

WHAT THE MODEL PRODUCED

Objective 1

Properties of Cylinders and Spheres

> EXPLANATION
> A **cylinder** is a solid with two parallel circular faces (the top and bottom) and one curved surface that connects them. The circular faces are congruent, and the line joining the centers of these faces is called the **axis** of the cylinder. The distance between the two faces is the **height** of the cylinder.
>
> A **sphere** is a perfectly round solid where every point on the surface is the same distance from the centre. It has only one continuous curved surface and no flat faces, edges or vertices.

> WORKED_EXAMPLE
> **Worked Example:**
>
> Identify the parts of a cylinder shown in a diagram.
>
> - The two flat circular faces are the **bases**.
> - The distance between the bases is the **height**.
> - The curved surface that wraps around the sides is the **lateral surface**.
>
> For a sphere, the only relevant part is its **surface**, which is the same everywhere, and the distance from the centre to any point on the surface is the **radius**.

1. How many circular faces does a solid cylinder have?
answer: 2

2. A cylinder has how many faces in total (including the curved surface as one face)?
answer: 3

3. How many flat faces does a sphere have?
answer: 0

4. What is the name of the line that joins the centres of the two circular faces of a cylinder?
answer: Axis

Objective 2

Finding the Volume of Cubes and Cuboids

> EXPLANATION
> The volume of a three-dimensional shape tells us how much space it occupies. For rectangular prisms (also called cuboids) you multiply the length, width and height:
>
> \[\text{Volume}=\text{length}\times\text{width}\times\text{height}\]
>
> All three dimensions must be in the same unit before you multiply.
>
> A cube is a special cuboid where all three edges are the same length. Its volume is therefore the edge length multiplied by itself three times:
>
> \[\text{Volume}=a\times a\times a = a^3\]
>
> Remember to keep the units consistent; the result will be in cubic units (cm³, m³, etc.).

> WORKED_EXAMPLE
> **Worked example 1:** Find the volume of a cuboid that is 8 cm long, 4 cm wide and 3 cm high.
>
> \[V = 8\,\text{cm} \times 4\,\text{cm} \times 3\,\text{cm} = 96\,\text{cm}^3\]
>
> **Worked example 2:** Find the volume of a cube with each edge 4 cm.
>
> \[V = 4\,\text{cm} \times 4\,\text{cm} \times 4\,\text{cm} = 64\,\text{cm}^3\]
>
> **Worked example 3:** A rectangular tank is 6 m long, 2 m wide and holds 36 m³ of water. How deep is the water?
>
> Let the depth be \(h\). Then
> \[6\,\text{m} \times 2\,\text{m} \times h = 36\,\text{m}^3\]
> \[12h = 36 \Rightarrow h = 3\,\text{m}\]

1. Calculate the volume of a cuboid 8 cm long, 4 cm wide and 3 cm high.
answer: 96 cm³

2. Calculate the volume of a cuboid with dimensions 5 m by 2 m by 3 m.
answer: 30 m³

3. What is the volume of a cube that measures 4 cm \(\times\) 4 cm \(\times\) 4 cm?
answer: 64 cm³

4. A rectangular tank 6 m long by 2 m wide holds 36 m³ of water. How deep is the water in the tank?
answer: 3 m

LAYER BY LAYER: WHAT WAS GROUNDED AND WHAT WAS NOT

Output layer	Made	Written from	Verdict
Title the reader-facing framing of the section	2	Scope	**grounded**
Explanation the principle, rule or relationship to review	2	Method written from event labels — the shape of the teaching without its substance	**substituted**
Worked example a problem with its solution worked through	2	nothing in the input	**unsourced**
Figure a drawing the learner reads from	0	nothing in the input	not produced

Practice what the learner does	8 Set work	**grounded**
Answer what the learner checks against	8 nothing in the input	**unsourced**